

HUTCHINSON, KS, USA

WMO: 724506

Lat: 38.065N

Lon: 97.861W

Elev: 1543

StdP: 13.89

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13986

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	10.2	-3.5	4.8	9.2	0.7	6.0	13.2	27.9	40.8	25.6	41.2	7.6	340	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	100.7	73.3	98.0	73.7	94.6	73.3	77.4	91.4	76.3	90.3	75.1	89.1	14.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	131.9	82.9	72.5	127.3	82.1	71.6	123.6	81.5	42.0	91.3	40.7	90.1	39.6	89.4	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.3	24.6	21.4	-3.7	105.2	4.4	3.0	-6.9	107.4	-9.4	109.1	-11.9	110.8	-15.0	113.0	
			DB												
			WB	-4.3	79.8	4.2	1.9	-7.4	81.2	-9.8	82.3	-12.2	83.3	-15.2	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	32.2	35.6	46.0	55.7	65.9	76.1	80.7	78.7	70.7	57.6	44.6	33.6	
	DBStd	19.47	10.65	11.22	10.78	9.56	8.39	6.83	6.22	6.04	8.16	9.55	9.77	10.09	
	HDD50	1988	554	416	202	50	2	0	0	0	0	37	215	512	
	HDD65	4717	1016	823	591	302	95	6	0	0	36	264	612	972	
	CDD50	4392	3	13	79	221	496	783	951	891	622	274	54	5	
	CDD65	1648	0	0	4	23	125	338	487	427	207	36	1	0	
	CDH74	19058	0	3	59	313	1310	3876	5956	4856	2242	421	22	0	
	CDH80	9478	0	0	9	84	494	1908	3302	2548	1004	127	2	0	
Wind	WSAvg	9.9	9.4	9.8	11.2	12.1	10.5	10.6	9.1	8.3	9.1	9.8	9.9	9.5	
Precipitation	PrecAvg	30.4	0.7	1.1	2.0	2.6	4.7	4.6	3.6	3.5	2.7	2.2	1.5	1.1	
	PrecMax	42.6	2.2	3.1	5.4	5.9	12.4	8.7	8.8	6.2	5.0	6.5	5.8	4.4	
	PrecMin	17.5	0.1	0.0	0.0	0.9	0.6	1.1	0.6	0.0	0.6	0.0	0.1	0.1	
	PrecStd	5.9	0.6	0.9	1.5	1.3	2.8	2.0	2.1	1.7	1.3	1.8	1.4	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	72.1	80.3	88.1	94.6	101.7	105.9	104.0	99.3	89.1	77.0	65.5	
		MCWB	48.9	52.7	58.6	65.4	70.1	73.1	73.4	73.5	72.1	67.3	57.7	52.3	
	2%	DB	60.2	64.9	74.4	81.6	89.8	97.1	101.8	99.9	93.3	83.2	71.8	59.5	
		MCWB	46.1	50.5	58.1	63.0	69.0	73.9	73.4	73.3	71.9	64.5	56.1	48.5	
	5%	DB	54.5	59.5	69.9	77.1	85.3	93.6	98.8	96.8	90.2	78.8	65.9	54.6	
		MCWB	43.3	47.8	55.6	61.2	68.3	72.9	74.0	73.7	71.0	62.7	53.0	45.4	
	10%	DB	49.4	54.1	64.3	72.7	81.5	90.6	94.9	92.6	86.1	73.5	61.3	50.0	
		MCWB	41.1	45.3	53.4	59.1	67.3	72.2	74.0	73.1	69.2	61.2	52.2	42.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	57.7	65.6	68.9	74.8	78.3	79.6	78.4	76.2	71.9	61.7	58.0
			MCDB	61.9	64.2	75.8	80.0	86.3	93.3	91.1	92.0	89.3	80.5	71.0	62.0
2%		WB	47.3	53.2	61.9	66.4	72.6	76.5	77.7	76.9	74.1	68.9	59.0	50.6	
		MCDB	56.5	60.1	68.9	76.3	84.0	90.7	91.4	91.0	87.3	77.5	65.8	55.9	
5%		WB	44.3	49.3	58.8	63.9	70.7	75.0	76.4	75.5	72.6	66.2	56.2	46.2	
		MCDB	53.7	59.9	66.0	73.5	81.5	88.9	90.6	89.5	85.4	74.6	63.9	53.5	
10%		WB	41.2	44.9	54.2	61.3	68.7	73.6	75.3	74.4	71.1	63.1	53.1	42.6	
		MCDB	49.6	53.3	63.5	71.4	78.7	86.9	89.5	88.1	83.3	71.1	60.5	49.8	
Mean Daily Temperature Range	5% DB	MDBR	22.8	22.9	24.4	24.4	23.5	23.3	23.5	23.5	24.4	24.8	24.1	21.8	
		MCDBR	32.0	32.3	32.5	30.1	27.8	26.0	28.1	28.6	27.4	30.3	31.8	29.6	
		MCWBR	20.1	20.3	18.2	16.1	12.2	8.9	7.6	8.8	10.3	14.9	18.0	19.5	
	5% WB	MCDBR	30.1	29.5	25.3	26.3	23.8	24.2	23.5	23.5	23.9	23.3	25.8	26.6	
		MCWBR	19.6	19.8	16.4	16.1	12.3	10.1	8.8	9.2	10.0	13.5	16.6	19.0	
Clear-Sky Solar Irradiance	taub	0.282	0.291	0.327	0.361	0.389	0.407	0.416	0.415	0.372	0.335	0.295	0.278		
	taud	2.511	2.492	2.416	2.347	2.345	2.321	2.332	2.310	2.393	2.477	2.518	2.519		
	Ebn at Noon	289	300	297	291	283	277	273	271	278	279	281	283		
	Edh at Noon	24	28	34	39	40	41	40	40	35	29	24	23		
All-Sky Solar Radiation	RadAvg	793	1031	1361	1708	1943	2194	2152	1925	1591	1172	870	687		
	RadStd	76	134	98	105	155	137	126	106	101	128	83	64		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page